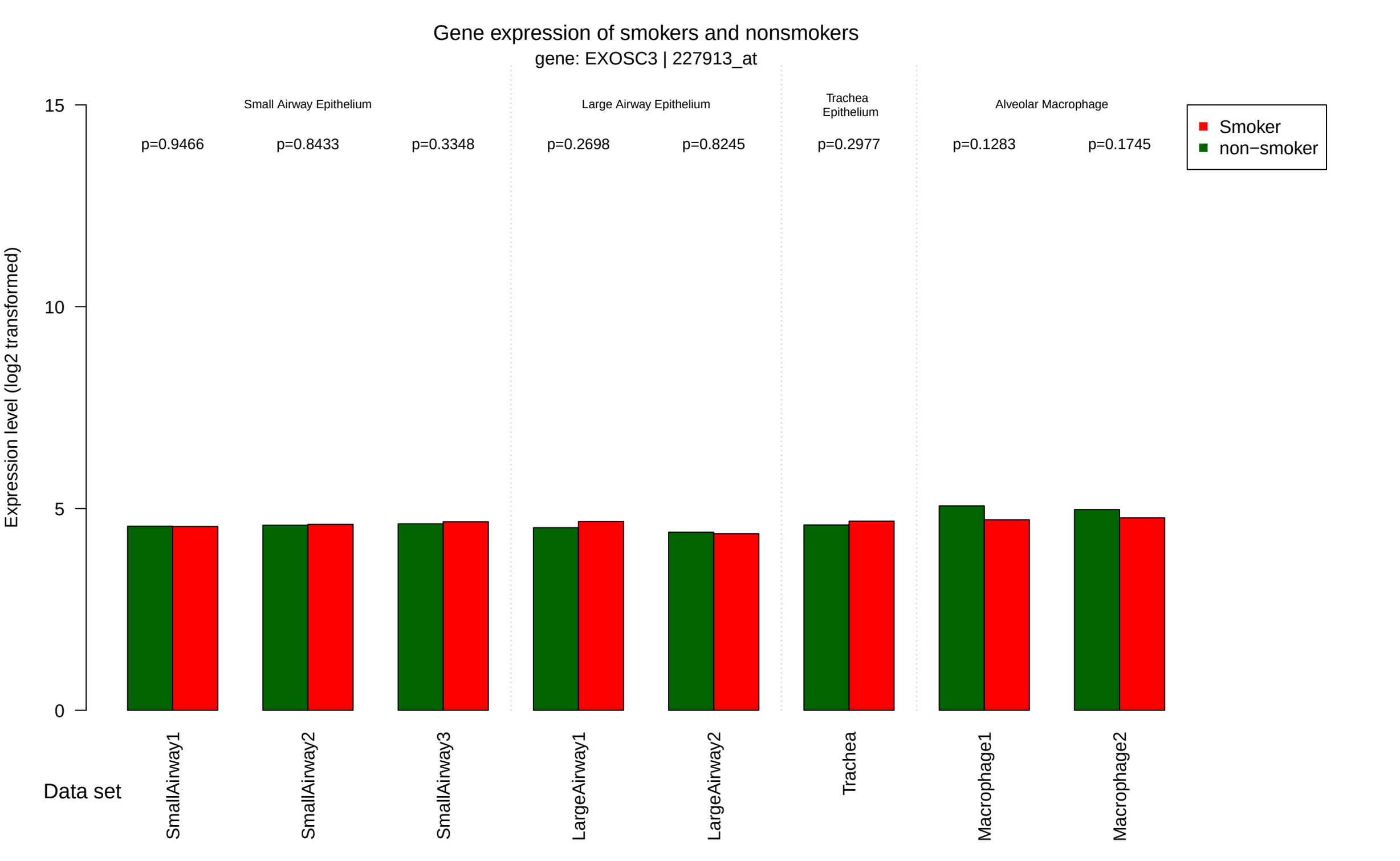
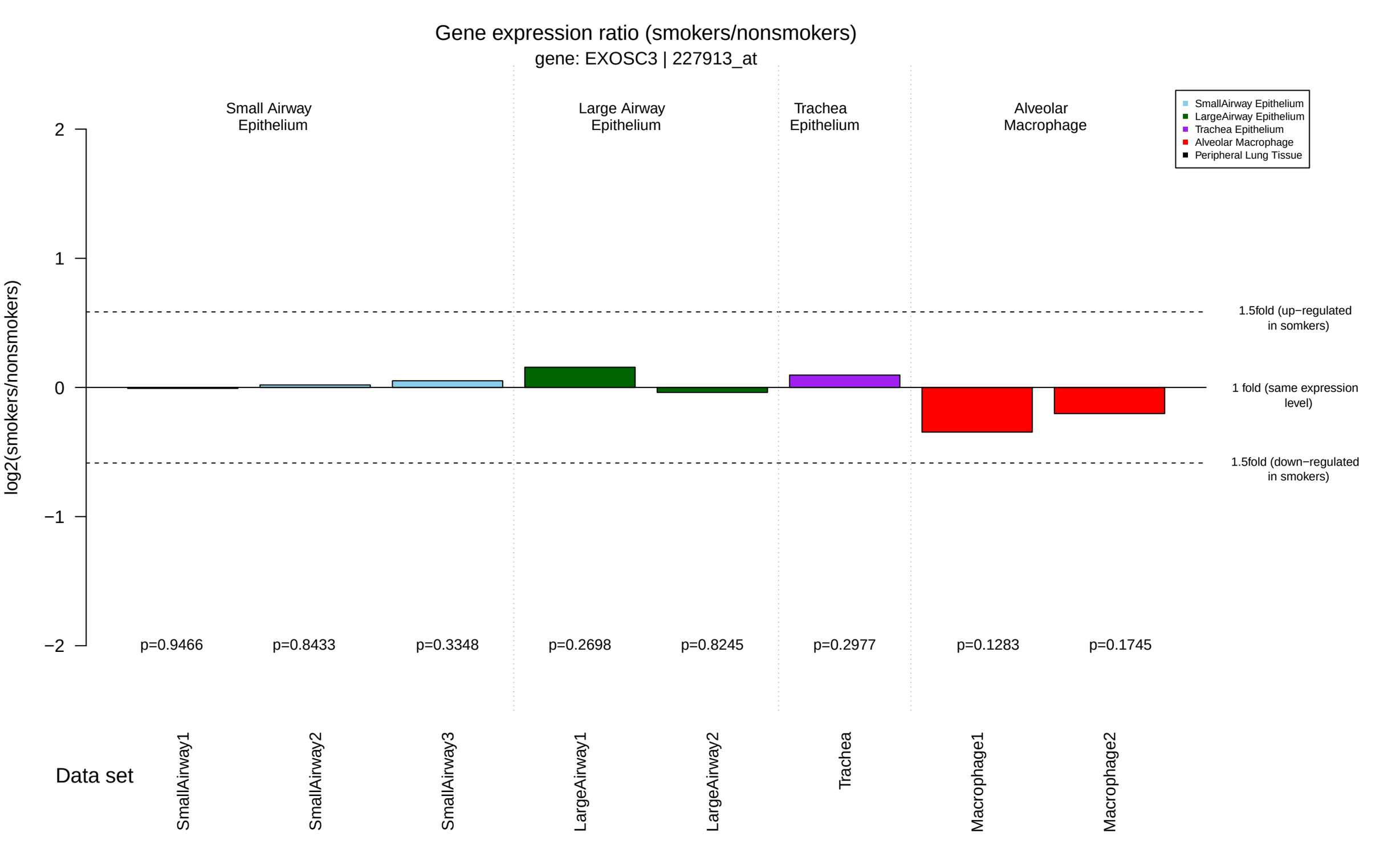
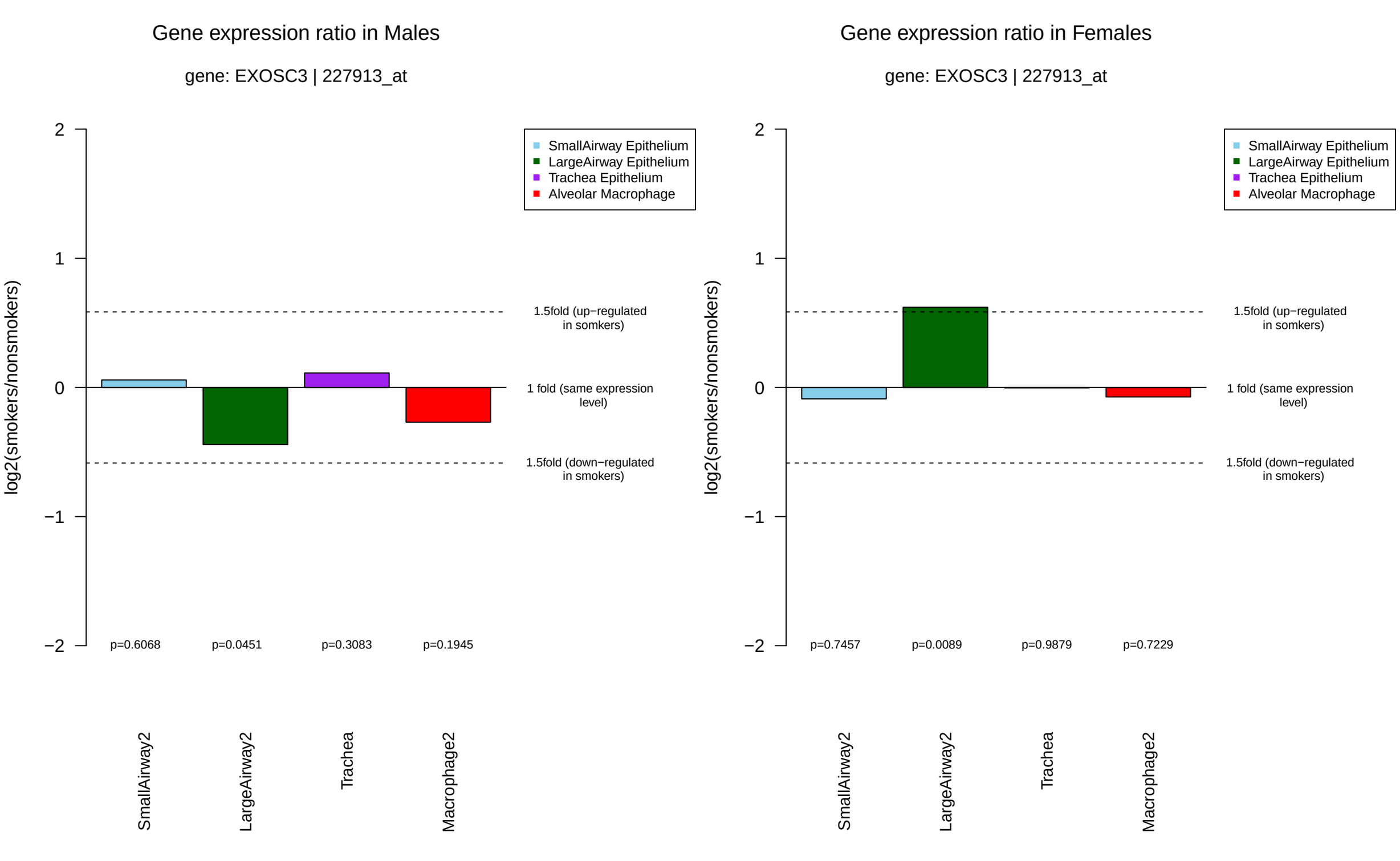
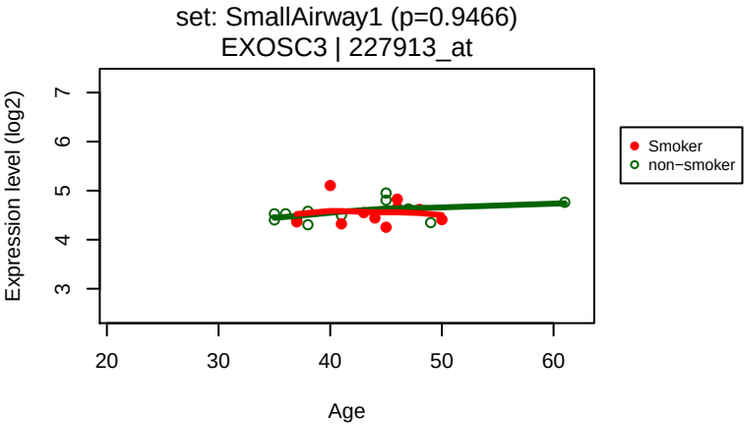
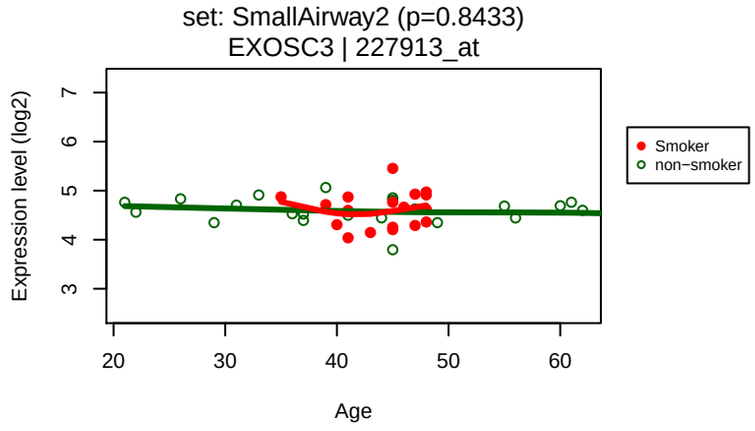
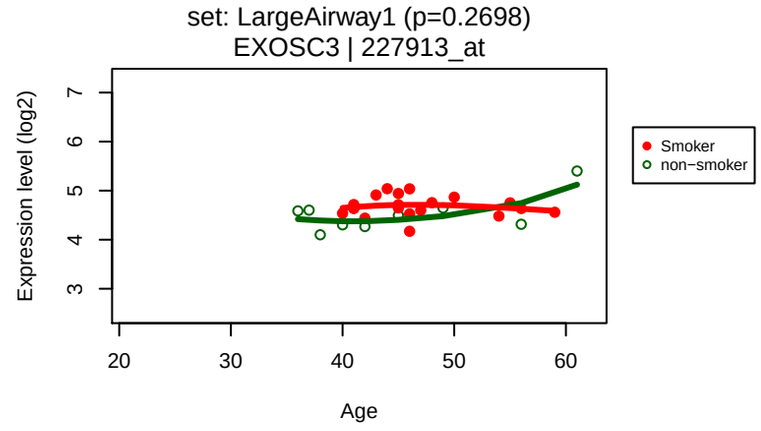

Gene expression of smokers and nonsmokers
gene: EXOSC3 | 227913_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9466
p=0.8433
p=0.3348
p=0.2698
p=0.8245
p=0.2977
p=0.1283
p=0.1745
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

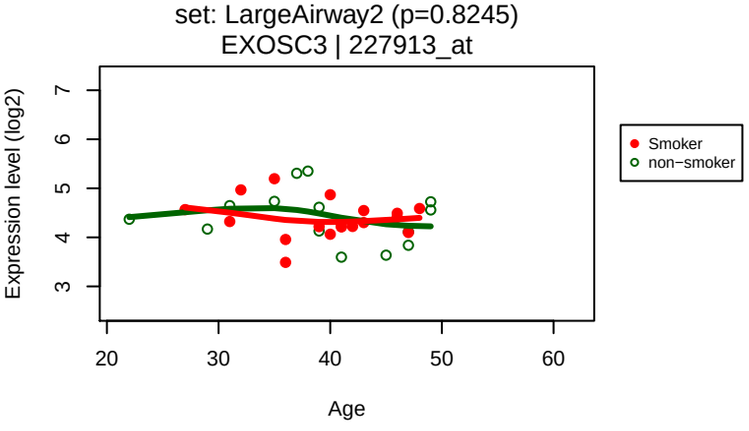

Gene expression ratio (smokers/nonsmokers)
gene: EXOSC3 | 227913_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9466
p=0.8433
p=0.3348
p=0.2698
p=0.8245
p=0.2977
p=0.1283
p=0.1745
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

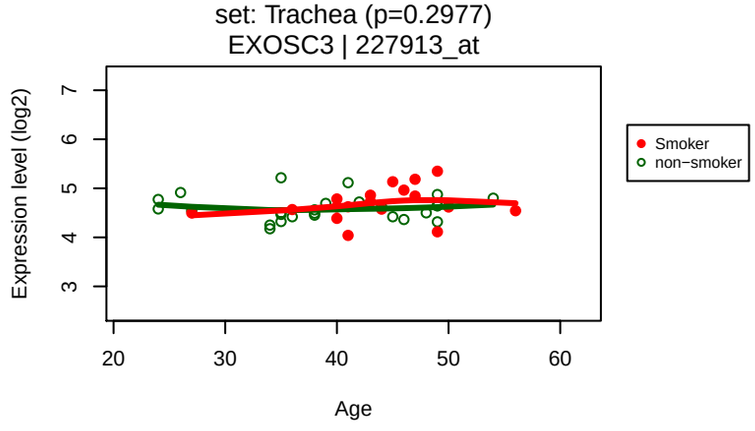

Gene expression ratio in Males
gene: EXOSC3 | 227913_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6068
p=0.0451
p=0.3083
p=0.1945
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: EXOSC3 | 227913_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7457
p=0.0089
p=0.9879
p=0.7229
SmallAirway2
LargeAirway2
Trachea
Macrophage2

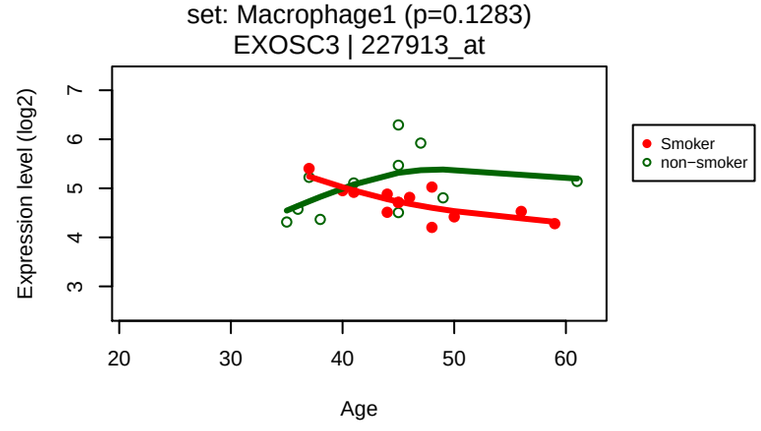

set: SmallAirway1 (p=0.9466)
EXOSC3 | 227913_at
Expression level (log2)
Age
Smoker
non−smoker

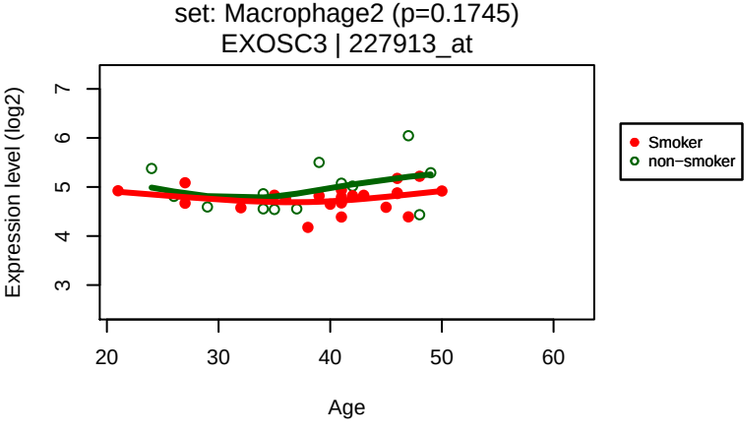

set: SmallAirway2 (p=0.8433)
EXOSC3 | 227913_at
Expression level (log2)
Age
Smoker
non−smoker

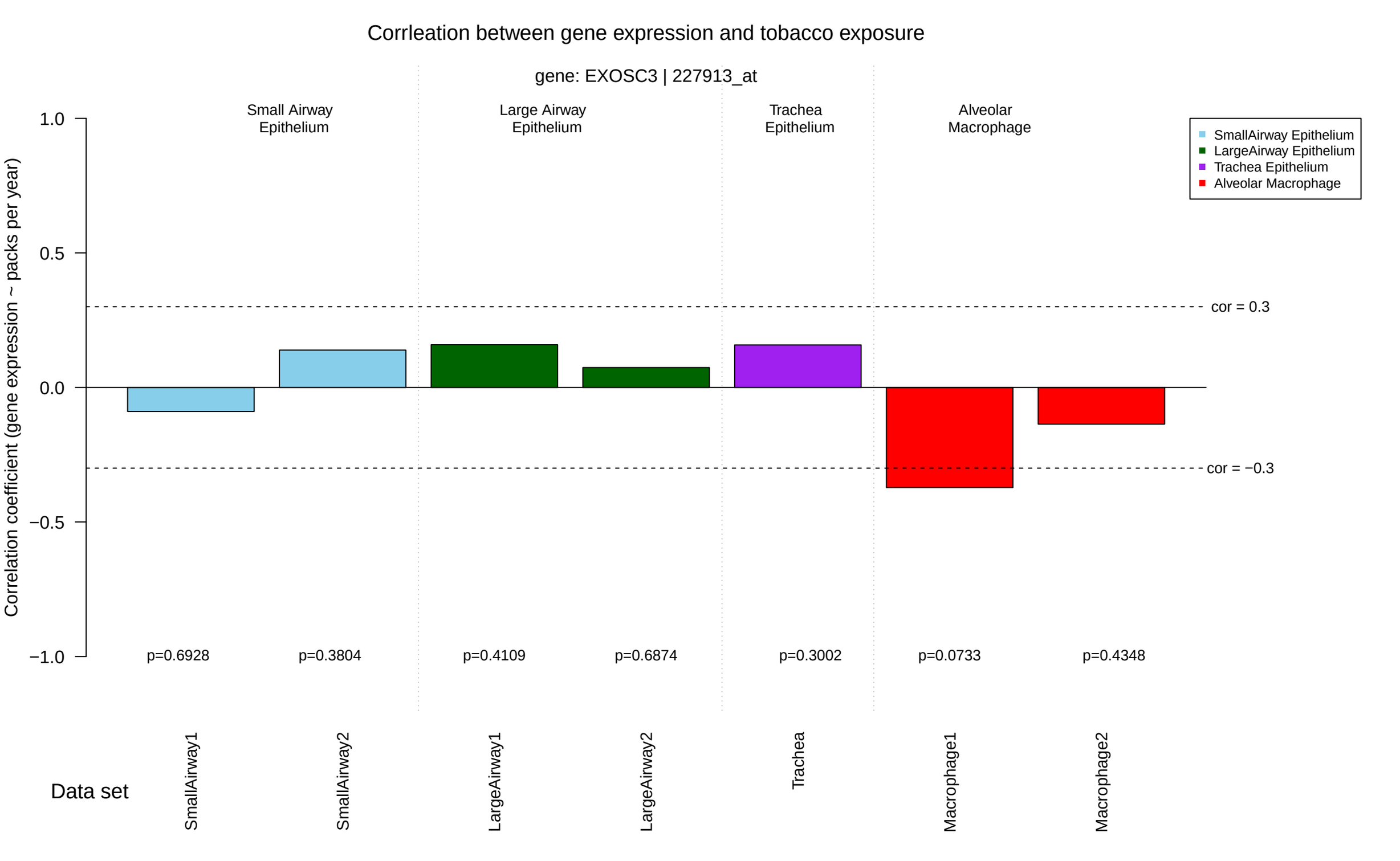

set: LargeAirway1 (p=0.2698)
EXOSC3 | 227913_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8245)
EXOSC3 | 227913_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2977)
EXOSC3 | 227913_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1283)
EXOSC3 | 227913_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1745)
EXOSC3 | 227913_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: EXOSC3 | 227913_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6928
p=0.3804
p=0.4109
p=0.6874
p=0.3002
p=0.0733
p=0.4348
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2